Karan Jadhav

Python Backend Engineer | FastAPI, Django, PostgreSQL, AWS

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Summary

Python backend engineer with 5+ years building production APIs, database-backed services, and asynchronous workflows. Strong in FastAPI, Django, PostgreSQL, service reliability, and translating data-heavy product requirements into maintainable backend systems.

Experience

Software Engineer (Backend & Data Infrastructure) - Intensel Ltd | March 2021 - May 2026

- Owned design and delivery of Python backend services for a climate-risk SaaS platform serving enterprise customers across 10+ countries, from requirement discovery through production support.
- Built FastAPI and Django REST APIs with request validation, pagination, caching, automated tests, and structured error handling; kept key customer-facing endpoints below 200 ms p99 in production.
- Designed service and data-access layers over PostgreSQL, PostGIS, and Redis, including schema changes, spatial indexes, query-plan analysis, and cache-aside behavior.
- Reduced critical building-footprint query paths over a 2.3B-record, 1.8 TB dataset from minutes to single-digit milliseconds by redesigning access paths and adding targeted indexes and caching.
- Built an LLM-powered agent that converted natural-language user requests into structured geospatial and analytics queries, using DuckDB to analyze geospatial vector datasets and integrating with backend APIs and PostgreSQL/PostGIS services.
- Built and operated a near-real-time pipeline that retrieved half-hourly NASA GPM IMERG precipitation GeoTIFFs with a 10-minute availability buffer and stored them in Amazon S3 for insurance and financial-services risk monitoring and alerts.
- Operated AWS SQS and Dask workflows processing 50K+ geospatial jobs per day with scheduling, retries, worker orchestration, and failure recovery.
- Built Docker-based CI/CD and deployed containerized services to Kubernetes on Amazon EKS, with CloudWatch observability across AWS EC2, S3, and RDS; reduced deployment time by 60% and improved release safety and incident diagnosis.
- Reviewed code, mentored engineers, and investigated failures across services, workers, queues, databases, and deployment environments.

Selected Projects

Global Building Footprints

- Built Python ingestion and query services that converted external building datasets into validated, indexed data for enterprise analytics.
- Separated high-volume geospatial processing from customer request paths through explicit API, worker, and database boundaries.

Map Tile Service

- Built authenticated backend delivery for 5.3 TB of raster data with caching, tuned storage access, and production monitoring.

Open Source

RediServe | Rust, Axum, Tokio

- Built an asynchronous Redis HTTP API with connection pooling and explicit backend resource management.

Technical Skills

Languages: Python, SQL; working knowledge of Rust

Backend: FastAPI, Django, REST APIs, validation, pagination, caching, automated testing

Databases & Queues: PostgreSQL, PostGIS, Redis, DuckDB, AWS SQS, Dask

Cloud & Operations: AWS, Kubernetes, Amazon EKS, EC2, S3, RDS, CloudWatch, Docker, Linux, CI/CD

AI & Analytics: LLM agents, natural-language query interfaces, geospatial analytics

Engineering: API design, system design, code review, mentoring, monitoring, production debugging

Education

Bachelor of Computer Science - University of Mumbai | 2020